Adding, Subtracting and Multiplying Polynomials

Adding, subtracting and multiplying polynomials, from a worked pattern to independent fluency

The pattern to copy

- **Adding:** drop the brackets, then collect like terms. $(x^2+4x)+(3x^2-x)=x^2+3x^2+4x-x=4x^2+3x$.
- **Subtracting:** change *every* sign in the second bracket first, then collect. $(x^2+4x)-(3x^2-x)=x^2+4x-3x^2+x=-2x^2+5x$.
- **Multiplying:** every term of the first bracket meets every term of the second, then collect.
- Write each answer in standard form — highest power first, no like terms left.

1. Simplify: $(3x^2+5x)+(2x^2-8x)$

Answer: ________________

2. Simplify: $(7x-4)-(2x-9)$

Answer: ________________

3. Multiply and simplify: $3x^2(4x-7)$

Answer: ____________________

4. Simplify: $(x^3-2x+6)+(4x^2+2x-11)$

Answer: ____________________

5. Multiply and simplify: $(x+9)(x-3)$

Answer: ____________________

6. Simplify: $(5x^2-x+3)-(x^2+4x-8)$

Answer: ____________________

7. Multiply and simplify: $(2x-5)(3x+4)$

Answer: ________________

8. Add all three polynomials and write the result in standard form:

$$(4x^3 - x) \; + \; (x^3 + 6x^2) \; + \; (-2x^3 + x - 9)$$

Answer: ________________

9. Subtract $(2x^2 - 7x + 1)$ from $(6x^2 + x - 4)$. Read the order carefully: the polynomial that comes after "from" is the one you start with.

Answer: ________________

10. Multiply and simplify: $(x+4)(x^2-3x+2)$

Each of the two terms in the first bracket has to meet all three terms in the second, so expect six products before you collect like terms.

Answer: ________________

11. Simplify: $(2x+3)^2-(x-1)(x+1)$

Expand each product first, then subtract — and remember that the whole second product is being subtracted.

Answer: ________________

12. A rectangle has length $(2x + 5)$ and width $(x + 3)$, both measured in the same unit.

(a) Write the perimeter in simplest form.

Answer: ____________________

(b) Write the area in standard form.

Answer: ____________________

(c) The perimeter has degree 1 but the area has degree 2. Explain why adding side lengths keeps the degree the same while multiplying them raises it.

Answer: ____________________